

Microeconomics

Lecture 8: Production Costs and Cost Curves

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What we know so far

- We have studied **technology**: $Q = f(L, K)$, isoquants, *MRTS*
- We have seen **returns to scale**: increasing, constant, decreasing

Next step

Just as the budget constrains the consumer, costs constrain the firm. The cost structure determines *pricing*, optimal scale, profitability.

Learning objectives

After this lecture you will be able to:

1. Distinguish the **short run** from the **long run**
2. Define and compute FC , $VC(Q)$, $TC(Q)$
3. Derive AC , AFC , AVC , MC from a cost function
4. Explain why MC intersects AC and AVC at their **minima**
5. Interpret cost curves in real business settings
6. Link AC to economies/diseconomies of scale

Focus of the lecture

Technology + input prices $\Rightarrow$ **cost function**.

Short run vs long run

Short run

- At least one input is **fixed** (typically K)
- The firm cannot change its production capacity
- Example: a pasta maker cannot buy new plants in a few days

Long run

- **All** inputs are variable
- The firm can freely choose its scale
- Example: the same pasta maker can build a new factory

Important note

It is not a matter of “3 months vs 3 years”. The long run is the time needed to vary *all* inputs. It depends on the industry.

Fixed, variable and total costs

$$TC(Q) = FC + VC(Q)$$

- **Fixed costs** FC : do not depend on Q
 - ▶ Rent, depreciation of machinery, management salaries
 - ▶ They exist even at $Q = 0$
- **Variable costs** $VC(Q)$: increase with Q
 - ▶ Raw materials, energy, direct labor
- **Total costs** $TC(Q)$: the sum

Example: Barilla

FC : Pedrignano plant (capacity 1,000 t/day), depreciation, fixed energy contracts.

VC : durum wheat, water, variable energy, on-call labor.

Average costs

$$AC(Q) = \frac{TC(Q)}{Q}, \quad AFC(Q) = \frac{FC}{Q}, \quad AVC(Q) = \frac{VC(Q)}{Q}$$

- $AC = AFC + AVC$ (always)
- AFC : *decreasing hyperbola*—the more you produce, the more you “spread” fixed costs over each unit
- AVC : typically U-shaped (linked to marginal returns)
- AC : U-shaped, always above AVC

Intuition: AFC in a startup

A startup with fixed costs (servers, licenses, offices) of €100,000/month: if it serves 1,000 users, $AFC = €100/\text{user}$; if it serves 100,000, $AFC = €1/\text{user}$. This is why digital platforms aim for scale.

Marginal cost MC

$$MC(Q) = \frac{dTC(Q)}{dQ} = \frac{dVC(Q)}{dQ}$$

Interpretation: additional cost of producing *one more unit*.

Link to marginal productivity:

- If I add one worker, I produce MPL more units
- Each worker costs w
- So $MC \approx \frac{w}{MPL}$: when MPL falls (diminishing returns), MC rises

Example: Ferrari

Each car requires hundreds of hours of craft work: adding one car to the production schedule means overtime and bottlenecks in specialized departments, so MC rises rapidly. Ferrari deliberately chooses not to expand production beyond the threshold at which MC starts to rise sharply, thus keeping average costs under control and preserving product scarcity.

Why MC intersects AC at its minimum

- If $MC < AC \Rightarrow$ producing one more unit lowers the average $\Rightarrow AC$ **falls**
- If $MC > AC \Rightarrow$ producing one more unit raises the average $\Rightarrow AC$ **rises**
- At the **minimum** of AC : $MC = AC$

The same logic applies to MC and AVC

Analogy: grade point average

Your average is 27/30. If the next grade is 24 (below the average), the average falls. If it is 30, the average rises. When they coincide, the average is at its minimum/maximum.

Exercise: the MC vs AC rule

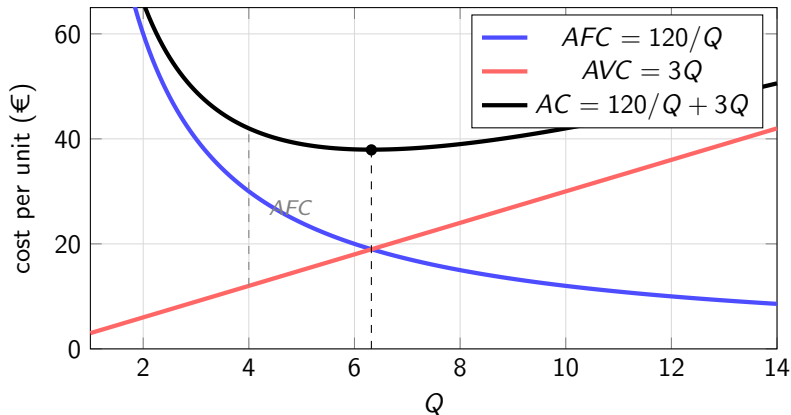
True or False? Justify in two lines.

1. If $MC < AC$, then AC is decreasing.
2. MC intersects AC at the maximum of AC .
3. MC does not depend on FC .
4. If $MC < AVC$, then AVC is decreasing.

Solution

1. **True:** the “next cost” is below the average $\Rightarrow$ it lowers the average.
2. **False:** MC intersects AC at its **minimum**, not its maximum.
3. **True:** $MC = \frac{dVC}{dQ}$; fixed costs do not vary at the margin.
4. **True:** the same averaging logic applies identically to AVC .

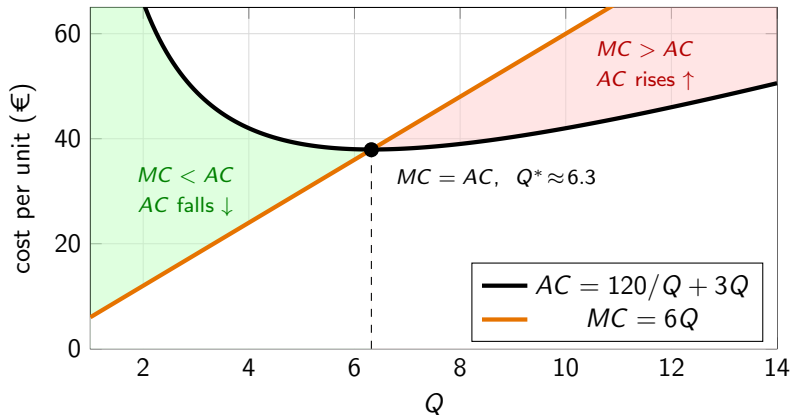
How average costs are built up



AFC always falls (fixed costs are spread over more units); **AVC** rises (diminishing returns); **AC** is their sum and is U-shaped.

Think of the startup in the previous example: the rent is already paid (FC); the relevant question is how much raw materials and labor cost per unit produced—and that is exactly what *AVC* shows us.

The $MC = AC$ rule at the minimum



While $MC < AC$, producing more *lowers* the average. Once $MC > AC$, it *raises* it. The minimum of AC is exactly where $MC = AC$.

Same logic as a grade average: while the next grade is below the average, the average falls; once above it, the average rises. The minimum of AC is the break-even point—the price below which the firm cannot cover all costs (fixed included). More on this with perfect competition.

Exercise: from TC to all the curves

Given:

$$TC(Q) = 120 + 3Q^2$$

1. Identify FC and $VC(Q)$.
2. Compute $AFC(Q)$, $AVC(Q)$, $AC(Q)$, $MC(Q)$.
3. Find the level Q^* that minimizes AC .
4. Verify that $MC(Q^*) = AC(Q^*)$.

Solution

$$FC = 120, \quad VC(Q) = 3Q^2$$

$$AFC(Q) = \frac{120}{Q}, \quad AVC(Q) = \frac{3Q^2}{Q} = 3Q$$

$$AC(Q) = \frac{TC}{Q} = \frac{120}{Q} + 3Q$$

$$MC(Q) = \frac{dTC}{dQ} = 6Q$$

Note: AFC is a hyperbola, AVC is linear and increasing, AC is U-shaped, MC is linear with twice the slope of AVC .

Solution: minimum of AC

$$\frac{dAC}{dQ} = -\frac{120}{Q^2} + 3 = 0$$

$$\frac{120}{Q^2} = 3 \quad \Rightarrow \quad Q^2 = 40 \quad \Rightarrow \quad Q^* = \sqrt{40} \approx 6.32$$

Check: $MC(Q^*) = 6 \cdot \sqrt{40}$ and $AC(Q^*) = \frac{120}{\sqrt{40}} + 3\sqrt{40} = \frac{120 + 120}{\sqrt{40}} = \frac{240}{\sqrt{40}} = 6\sqrt{40}$. ✓

Interpretation: up to $Q^* \approx 6.32$ the “spreading fixed costs” effect dominates and AC falls; beyond Q^* the increase in variable costs dominates and AC rises.

Real-world example: the 2022 energy crisis in Emilia-Romagna

In 2022 the price of natural gas in Europe rose by more than 300%. Manufacturing firms in Emilia (ceramics, engineering, food) suffered a strong shock to variable costs.

Effects on the cost structure:

- Energy is a **variable** input $\Rightarrow$ shock to $VC(Q)$
- AVC and MC shift upward
- AFC remains unchanged
- The Q^* that minimizes AC shifts (the balance between FC and VC changes)

Discussion

Which firms suffered most? Energy-intensive ones (Sassuolo ceramics producers), those with low margins, and those operating in markets with elastic demand (they cannot pass the cost on to consumers).

Exercise: variable cost shock

Go back to $TC(Q) = 120 + 3Q^2$.

An energy shock raises variable costs by **20%**: new $VC'(Q) = 1.2 \cdot 3Q^2 = 3.6Q^2$.

1. Write the new $TC'(Q)$.
2. Compute $AVC'(Q)$ and $MC'(Q)$.
3. Does FC change?
4. Find the new Q^{**} that minimizes AC' . Is it larger or smaller than $Q^* \approx 6.32$?

Solution

$$TC'(Q) = 120 + 3.6Q^2$$

$$AVC'(Q) = 3.6Q, \quad MC'(Q) = 7.2Q$$

$FC = 120$ (unchanged—it does not depend on output)

Minimum of AC' :

$$\frac{dAC'}{dQ} = -\frac{120}{Q^2} + 3.6 = 0 \quad \Rightarrow \quad Q^{**} = \sqrt{\frac{120}{3.6}} = \sqrt{33.3} \approx 5.77$$

The minimum shifts to the left: with more expensive variable inputs, it pays to produce less to minimize average cost per unit.

Costs in the long run

In the long run all inputs are variable. The firm chooses the combination (L, K) that minimizes the cost of producing Q :

$$C(Q) = \min_{L, K} \{wL + rK : f(L, K) \geq Q\}$$

$$AC(Q) = \frac{C(Q)}{Q}$$

Relationship with returns to scale:

- Increasing RTS $\Rightarrow$ decreasing AC (economies of scale)
- Constant RTS $\Rightarrow$ constant AC
- Decreasing RTS $\Rightarrow$ increasing AC (diseconomies of scale)

Long-run average cost curve and minimum efficient scale

The long-run AC curve is typically U-shaped:

- At low quantities: economies of scale (fixed costs, specialization)
- At high quantities: diseconomies of scale (coordination, bureaucracy)
- The minimum point is called the **minimum efficient scale** (MES)

Short-run AC curves always lie above the long-run one (or touch it): the long-run AC curve is the *lower envelope* of the short-run AC curves.

Example: Novamont (bioplastics)

Initial investment in R&D and plants: very high FC , long-run AC initially high. As production grows, AC falls thanks to economies of scale. Reaching the MES is the strategic challenge in the early years.

Exercise: reasoning about long-run AC

For each of the following industries, do you expect a decreasing, constant or increasing AC (over the relevant range)? Briefly justify.

1. Microchip production (high investment in fabs, huge economies of scale)
2. Individual professional services (lawyers, accountants)
3. Large-scale organized retail (supermarket chains)

Solution

1. **Microchips:** strongly **decreasing** LAC —design and fab costs are huge but fixed; the marginal cost of producing one more chip is almost zero. Economies of scale dominate (TSMC, Intel).
2. **Individual professional services:** **increasing** or quickly constant LAC —the professional is the limiting factor; scaling up means hiring and coordinating others, with rising supervision costs and quality risk.
3. **Large-scale retail:** LAC initially **decreasing** (centralized purchasing, logistics, IT), then tends to level off or rise when the chain becomes too large to manage. Example: Coop vs local discount stores.

Exercise

Given $TC(Q) = 200 + 5Q + Q^2$:

1. Identify FC and $VC(Q)$.
2. Compute $AFC(Q)$, $AVC(Q)$, $AC(Q)$, $MC(Q)$.
3. What is the shape of AVC ? (linear, U-shaped, hyperbola?)
4. Find the Q^* that minimizes AC .

Solution

$$FC = 200, \quad VC(Q) = 5Q + Q^2$$

$$AFC = \frac{200}{Q}, \quad AVC = 5 + Q, \quad AC = \frac{200}{Q} + 5 + Q, \quad MC = 5 + 2Q$$

AVC is **linear and increasing** (not U-shaped in this case—variable costs are increasing from the start).

Minimum of AC :

$$\frac{dAC}{dQ} = -\frac{200}{Q^2} + 1 = 0 \quad \Rightarrow \quad Q^* = \sqrt{200} \approx 14.14$$

Check: $MC(Q^*) = 5 + 2\sqrt{200}$; $AC(Q^*) = \frac{200}{\sqrt{200}} + 5 + \sqrt{200} = \sqrt{200} + 5 + \sqrt{200} = 5 + 2\sqrt{200}$. ✓

Exercise: minimum of AC

Consider a generic cost function:

$$TC(Q) = F + cQ^2 \quad (F > 0, c > 0)$$

1. Find, in symbolic form, the Q^* that minimizes AC .
2. How does Q^* change if F increases? And if c increases?
3. Give an economic interpretation of the results.

Solution

$$AC(Q) = \frac{F}{Q} + cQ \quad \Rightarrow \quad \frac{dAC}{dQ} = -\frac{F}{Q^2} + c = 0$$

$$Q^* = \sqrt{\frac{F}{c}}$$

Effect of $F \uparrow$: $Q^* = \sqrt{F/c}$ increases—higher fixed costs make it optimal to produce more to spread them.

Effect of $c \uparrow$: Q^* decreases—more expensive variable inputs make it optimal to produce less.

Interpretation

Capital-intensive firms (high F) have a higher minimum efficient scale. Firms with costly variable inputs (high c) are better kept small.

Case Study: Amazon Web Services

The context

AWS provides cloud services to millions of companies worldwide. In 2023 it had \$91 billion in revenue with operating margins above 25%.

Cost structure

- Huge FC : data centers, physical infrastructure, R&D
- Near-zero VC for each additional customer
- Prices offered: often lower than those of smaller competitors

Questions for the group

1. How does AFC behave as the number of customers grows? Sketch its path.
2. How high is MC ? Why?
3. How can AWS charge low prices while remaining profitable?
4. What would happen to the cost structure if the number of customers halved?

Time: 8–10 minutes. Appoint a spokesperson to report back.

Debrief: AWS Case Study

1. $AFC = FC/Q$ collapses rapidly as customers increase (hyperbola): with hundreds of millions of users the fixed cost per customer becomes negligible.
2. $MC \approx 0$: activating a new virtual server has an almost zero marginal cost. Variable costs are negligible relative to fixed costs.
3. Economies of scale allow huge FC to be spread over a huge customer base, lowering AC to levels that smaller competitors cannot reach. The competitive advantage *is* scale.
4. With half the customers: AFC doubles $\Rightarrow AC$ rises sharply $\Rightarrow$ margins erode. A model built on high FC and low VC is sustainable **only** at large scale.

Link to theory

This is an extreme case of $Q^* = \sqrt{F/c}$: with very high F and $c \approx 0$, the minimum efficient scale is huge. Those who do not reach it cannot compete on price.

Overall summary

- $TC = FC + VC(Q)$
- $AC = AFC + AVC$
- $MC = dTC/dQ = dVC/dQ$
- MC intersects AC and AVC at their **minima**
- LAC : envelope of the SR curves
- Decreasing $LAC \Leftrightarrow$ economies of scale

Skills acquired

- ✓ Go from TC to all the cost curves
- ✓ Find the Q^* that minimizes AC
- ✓ Interpret shocks to FC and VC
- ✓ Read graphs and draw implications
- ✓ Link cost structure to business decisions

Optional homework exercises

1. With $TC = 50 + Q^2$, compute AC , AVC , MC and find Q^* .
2. Explain why AFC is always decreasing.
3. For what type of industry would you expect a very high MES?